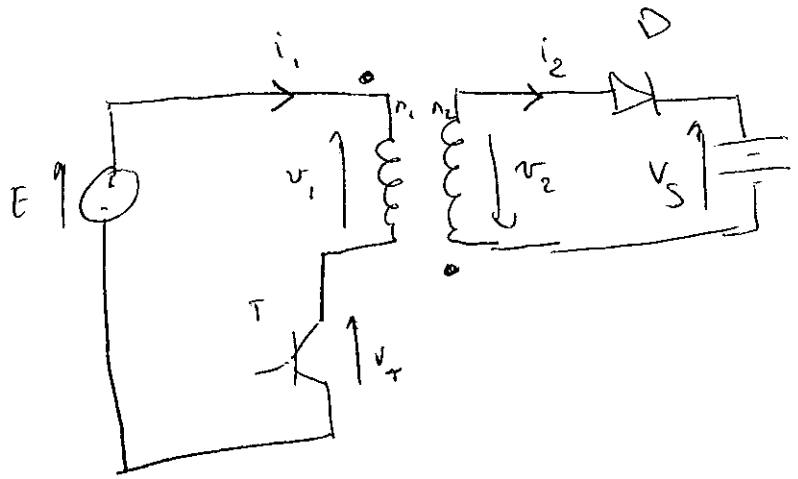
Flyback Switching Power Supply for Battery Charger

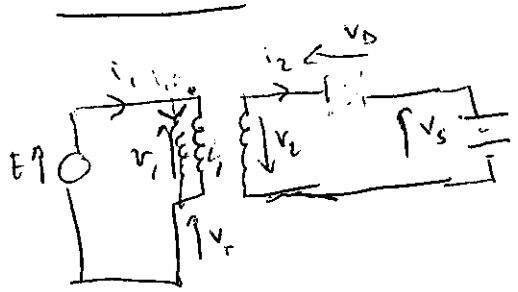


$V_S = 24V$

$E = 300V.$

1.1) $n_1 i_1 + n_2 i_2 = \mathcal{R}\varphi$

1.2) $0 < t < \alpha T$: T "closed" $\rightarrow \boxed{V_T = 0}$



$\boxed{v_1 = E} = n_1 \frac{d\varphi}{dt}$

$v_2 = \frac{n_2}{n_1} v_1 > 0$

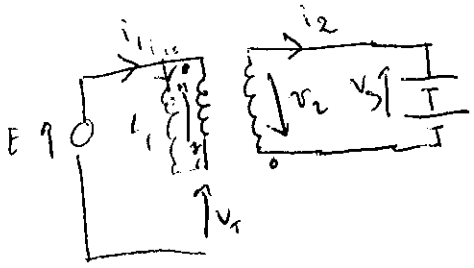
$V_D = -v_2 - V_S < 0 \rightarrow$ D "reverse"

$\rightarrow \boxed{i_2 = 0}$

$\varphi = \frac{n_1}{\mathcal{R}} i_1$

$v_1 = L_1 \frac{di_1}{dt}$ and boundary mode $\varphi(t=0) = 0 \rightarrow i_1(t=0) = 0.$

$\alpha T < t < T$: T "open" $\rightarrow \boxed{i_1 = 0} \rightarrow$ flux continuity -> D "forward"



$v_2 = -V_S$

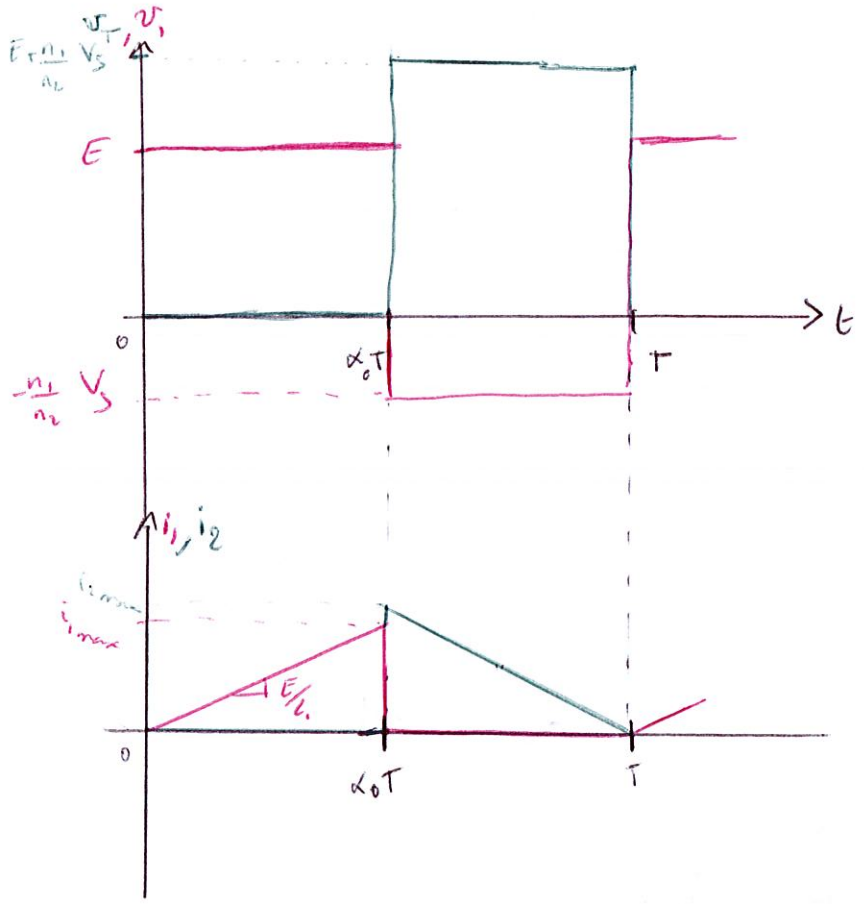
$\boxed{v_1 = -\frac{n_1}{n_2} V_S}$

$\boxed{v_T = E + \frac{n_1}{n_2} V_S} > 0$

$n_2 i_2 = \mathcal{R}\varphi$ et $v_2 = -V_S = n_2 \frac{d\varphi}{dt}$

$\rightarrow \frac{di_2}{dt} = \frac{\mathcal{R}}{n_2} \frac{d\varphi}{dt} = -\frac{\mathcal{R}}{n_2^2} V_S$

$\rightarrow$boundary mode : $i_2(t=T) = 0$



1.2) $\langle v_1 \rangle = 0 \Rightarrow E\alpha_0 - \frac{n_1}{n_2} V_{S_0}(1-\alpha_0) = 0$

$$\Rightarrow V_{S_0} = \frac{E\alpha_0}{(1-\alpha_0)\frac{n_1}{n_2}} = \frac{E m \alpha_0}{1-\alpha_0}$$

$$\Rightarrow m = \frac{V_{S_0}(1-\alpha_0)}{E\alpha_0} = 0{,}067.$$

1.3) $I_{S_0} = (1-\alpha_0)\cdot\frac{i_{2max}}{2} = \frac{1-\alpha_0}{2}\cdot\frac{\Re\varphi_{max}}{n_2} = \frac{1-\alpha_0}{2}\cdot\frac{n_1}{n_2}\cdot i_{1max}$

$$= \frac{1-\alpha_0}{2m}\cdot\frac{E}{L_1}\alpha_0 T = \frac{(1-\alpha_0)E\alpha_0}{2mL_1 f}$$

$$\Rightarrow L_1 = \frac{(1-\alpha_0)\alpha_0 E}{2mf I_{S_0}} = 2{,}8\,mH.$$

1.4) $W_D = E.\langle i_1 \rangle . T = E\alpha_0 . \frac{E}{2L_1}\alpha_0 T . T = \frac{E^2\alpha_0^2}{2L_1 f^2}$

$$W_D = V_S \langle i_2 \rangle . T = V_S . I_{S_0} . T = \frac{V_S I_{S_0}}{f}$$

1.5) $W_0 = \frac{V_{s_0} I_{s_0}}{f} = 0{,}01\,J$.

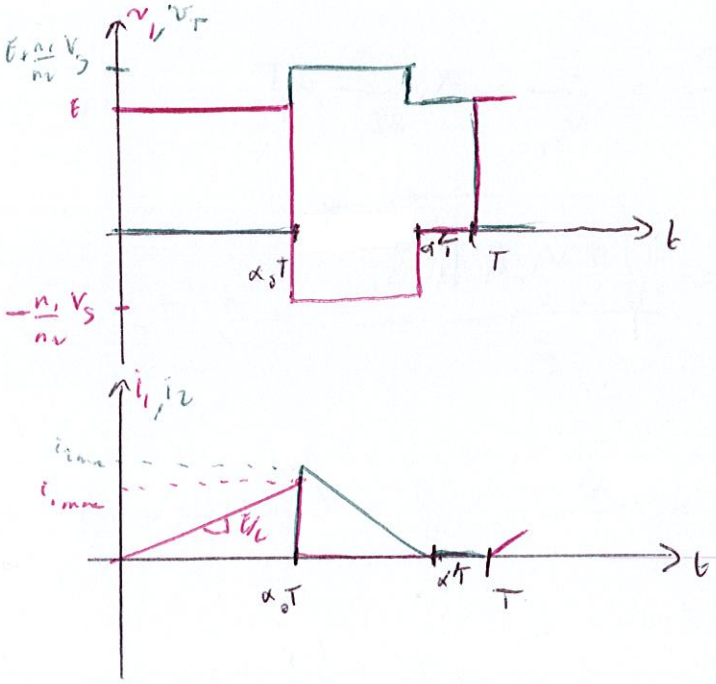
Batterie : 250 Ah : $W_{batt} = 250 \times 3600 \times \underset{\text{batterie 24V}}{24} = 21{,}6 \cdot 10^6\,J$

charging time : $\frac{\overset{W_{batt}}{21{,}6 \cdot 10^6}}{\underset{W_0}{0{,}01} \cdot \underset{f}{20\,000}} = 108\,000\,s = 30\,h$.

2.1) $V_s \nearrow$ donc $\alpha T < t < T$: $\frac{di_2}{dt} = -\frac{R\,V_s}{n_2^2}$

=> <u>discontinuous</u> <u>mode</u>

2.2)



$\alpha' T < t < T$: T open and D "reverse"

$i_1 = i_2 = 0 \qquad \varphi = 0$

$v_1 = n_1 \frac{d\varphi}{dt} = 0 = v_2$

$v_T = E$

2.3) $i_{1max} = \frac{E}{L_1} \alpha_0 T$

flux continuity at $t = \alpha_0 T$: $n_2 i_{2\,max} = n_1 i_{1max}$

$\Rightarrow i_{2\,max} = \frac{E \alpha_0 T}{L_1 m}$

Hopkinson : $n_1 i_1 + n_2 i_2 = n_1 i_{10}$

$\alpha_0 T < t < \alpha' T$: $i_1 = 0 \rightarrow n_2 i_2 = n_1 i_{10} \Rightarrow \frac{di_2}{dt} = \frac{1}{m} \frac{di_{10}}{dt} = \frac{1}{m L_1} v_1 = \frac{E + \frac{V_s}{m}}{m L_1}$

Time conduction of diode D

$$\Delta t = \frac{i_{2max}}{\frac{di_2}{dt}} = \frac{\frac{E\alpha_0 T}{\cancel{L_1 m}}}{\frac{E + \frac{V_S}{m}}{\cancel{m L_1}}}$$

$$\Delta t = \frac{E\alpha_0 T}{E + \frac{V_S}{m}} = 11,4\ \mu s$$

It is possible to check that $\Delta t < \frac{T}{2} =$

2.4) $W_D = E\langle i_1 \rangle T = 0,01\ J$ (energy unchanged)

$$W_D = \frac{V_{S_1} I_S}{f} \Rightarrow I_S = \frac{W_D f}{V_{S_1}} = 8,3\ A.$$

2.5) $$I_{S_1} = \frac{W_D f}{V_{S_1}} = \frac{E\langle i_1 \rangle}{V_{S_1}} = \frac{E}{V_{S_1}} \cdot \alpha_1 \frac{E}{2L_1} \alpha_1 T$$

$$= \frac{E^2 \alpha_1^2}{2 V_{S_1} L_1 f} \Rightarrow \alpha_1 = \frac{\sqrt{2 V_{S_1} L_1 f I_{S_1}}}{E} = 0,17.$$

3.1) $$\varphi_{max} = A_e B_{max} = \frac{E}{n_1} \alpha_0 T \quad \leftarrow v_1 = n_1 \frac{d\varphi}{dt} = E \ (0 < t < \alpha_0 T)$$

$$\rightarrow B_{max} = \frac{E\alpha_0 T}{n_1 A_e}$$

3.2) $$A_{e_{min}} = \frac{E\alpha_0 T}{n_1 B_{sat}}$$

4.1) $$i_{1max} = \frac{2 m I_S}{1-\alpha} \quad (\text{Cf Q 1.3})$$

$$I_{1eff}^2 = \frac{1}{T}\int_0^t i_1^2(t)\,dt = \frac{1}{T}\int_0^{\alpha T} \left(\frac{i_{1max}\, t}{\alpha T}\right)^2 dt = \frac{\cancel{T}^2 i_{1max}}{\alpha^2 T^3} \int_0^{\alpha T} t^2\,dt = \frac{i_{1max}^2}{\cancel{\alpha^2 T^3}} \cdot \frac{\cancel{\alpha^3 T^3}}{3}$$

$$I_{1eff}^2 = i_{1max}^2 \frac{\alpha}{3} \Rightarrow I_{1eff} = i_{1max}\sqrt{\frac{\alpha}{3}} = \frac{2 m I_S}{1-\alpha}\sqrt{\frac{\alpha}{3}}$$

4.2) $I_{2eff} = I_{2max}\sqrt{\frac{1-\alpha}{3}}$ (same calculus)

$$= \frac{i_{1max}}{m}\sqrt{\frac{1-\alpha}{3}} = \frac{2I_s}{1-\alpha}\sqrt{\frac{1-\alpha}{3}}$$

4.3) $S_{TCU} = \frac{n_1 I_{1eff}}{\delta} + \frac{n_2 I_{2eff}}{\delta}$

$$= \frac{2n_2 I_s}{(1-\alpha)\delta}\sqrt{\frac{\alpha}{3}} + \frac{2n_2 I_s}{(1-\alpha)\delta}\sqrt{\frac{1-\alpha}{3}}$$

$$= \frac{2n_2 I_s}{(1-\alpha)\delta}\left(\sqrt{\frac{\alpha}{3}} + \sqrt{\frac{1-\alpha}{3}}\right)$$

4.4) $S_B = \frac{S_{TCU}}{k_B} = \frac{2n_2 I_s}{(1-\alpha)\delta k_B}\left(\sqrt{\frac{\alpha}{3}} + \sqrt{\frac{1-\alpha}{3}}\right)$

5.1) $A_c \cdot S_B > \frac{E\alpha_0 T}{n_1 B_{sat}} \cdot \frac{2n_2 I_s}{(1-\alpha)\delta k_B}\left(\sqrt{\frac{\alpha}{3}} + \sqrt{\frac{1-\alpha}{3}}\right)$

M

$P_s = V_s I_s = \frac{E_n \alpha}{1-\alpha} I_s \Rightarrow A_c S_B > \frac{2P_s T}{B_{sat}\delta k_B}\left(\sqrt{\frac{\alpha}{3}} + \sqrt{\frac{1-\alpha}{3}}\right)$

5.2) $A_c \cdot S_B = 16328\ mm^4$

5.3)

Noyau	A_c (mm²)	S_B (mm²)	A_c sans entrefer	A_c avec entrefer	$A_c \cdot S_B$ (mm⁴)
RM 6	31	32	3100	---	992
RM 8	55	57	4100	160	3135
RM 10	90	95	5500	250	8550
RM 12	125	128	6000	300	16 000
RM 14	170	176	6800	400	29 920

5.4) $B < B_{sat} = \frac{E \alpha_0 T}{n_1 A_e}$

$\Rightarrow n_1 < \frac{E \alpha_0 T}{B_{sat} A_e} = 110,3 \rightarrow \boxed{n_1 = 111 \text{ turns}}$

$m = \frac{n_2}{n_1} \rightarrow n_2 = m n_1 = 7,4 \quad \boxed{n_2 = 8 \text{ turns}}$

5.5) $L_1 = n_1^2 A_L \rightarrow A_L = \frac{L_1}{n_1^2} = 227 \text{ nH}.$

$\rightarrow$ Need an air gap, but 0.5mm is not enough